

Question	Marking Guidance	Mark	Comments
01.1	1. Replication of (circular) <u>DNA</u> ; 2. Replication of <u>plasmids</u> ; 3. (New) membrane forms (dividing cell) OR Division of cytoplasm (to produce daughter cells);	3 (3 x AO1)	1. and 2. Accept doubling for replication 1. Reject chromosome 1. Reject mitosis 1. Reject single strand of DNA 2. Ignore (named) organelles replicate 3. Accept descriptions of either alternative 3. Accept cytokinesis (occurs)

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01.2	Correct answer of 1159 = 2 marks;; Evidence of 1158 (incorrect rounding) = 1 mark OR Evidence of 1158 followed by any number of decimal places (not to nearest whole number) = 1 mark OR Evidence of 1259 (answer that failed to subtract 100) = 1 mark OR Evidence of 12 (not multiplied by 100) = 1 mark OR Evidence of 145 900(.0651) (calculation of increase in number of bacteria) = 1 mark OR Evidence of 92 (dividing by $10^{5.2}$ instead of $10^{4.1}$) = 1 mark OR Evidence of 27 (percentage increase of log values read from graph) = 1 mark OR Evidence of the following working (calculation method including correct powers of 10) = 1 mark; $\frac{10^{5.2} - 10^{4.1}}{10^{4.1}} \times 100$ OR $\left(\frac{10^{5.2}}{10^{4.1}} \times 100\right) - 100$ OR $\frac{158489(.32) - 12589(.25)}{12589(.25)} \times 100$ OR $\frac{145900(.07)}{12589(.25)} \times 100$ OR $\frac{158489(.32)}{12589(.25)}$	2 (2 x AO2)	

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01.3	1. Fewer/no protons/hydrogen ions/H⁺ diffuse/move through ATP synthase; 2. Less/no energy to add phosphate to ADP OR Less/no energy for phosphorylation of ADP OR Less/no energy to form ATP OR Less/no oxidative phosphorylation; 3. (Bacteria die due to) less/no ATP/energy for cellular/metabolic processes;	3 (3 x AO2)	1. Accept no/less chemiosmosis 1. Reject reference to protons/hydrogen ions/H⁺ diffusing if in mitochondrion 1. Accept hydrogen/H for protons/hydrogen ions/H⁺ (as told hydrogen ion in stem of question) 2. or 3. Reject less/no energy produced once 2. Accept (P), P_i or PO₄³⁻ for phosphate 2. Reject phosphorous/P 3. Accept named process e.g. synthesis of an organic molecule, binary fission, DNA replication, active transport, etc. 3. Reject 'less/no energy/ATP for respiration' unless qualified e.g. glucose to glucose phosphate 3. Reject incorrect metabolic process for a bacterium, e.g. muscle contraction
